

SIMATS ENGINEERING

ECA09 - DIGITAL SIGNAL PROCESSING LAB

Experiment 07.1

Power Spectral Density Estimation and Spectral Analysis of Noisy Sinusoidal Signals Using Periodogram Method in MATLAB.

Name: Dhinakaran. S **Reg. No.:** 192412353

Aim

To implement and analyze the experiment described in the official ECA09 MATLAB lab manual.

Procedure / Calculations

Follow the experiment workflow given in the official lab manual: initialize MATLAB, define the specified signal/filter parameters, execute the algorithm, plot the required responses, and record the result.

MATLAB Program

```
clc; clear; close all
fs=1000; t=(0:999)/fs;
s=sin(2*pi*100*t)+0.5*sin(2*pi*200*t);
n=0.3*randn(size(t));
x=s+n;
figure, plot(x), grid on
title('Signal with Noise')
figure, periodogram(x,[],[],fs)
title('PSD using Periodogram')
figure
X=abs(fft(x));
f=(0:length(X)-1)*fs/length(X);
plot(f(1:500),X(1:500)), grid on
title('Magnitude Spectrum')
figure
bar([mean(s.^2) var(n)])
set(gca,'XTickLabel',{'Signal','Noise'})
title('Power Comparison')
grid on
```

Output / Analysis

The output/analysis pages below are reproduced from the uploaded official ECA09 MATLAB lab manual for this experiment.

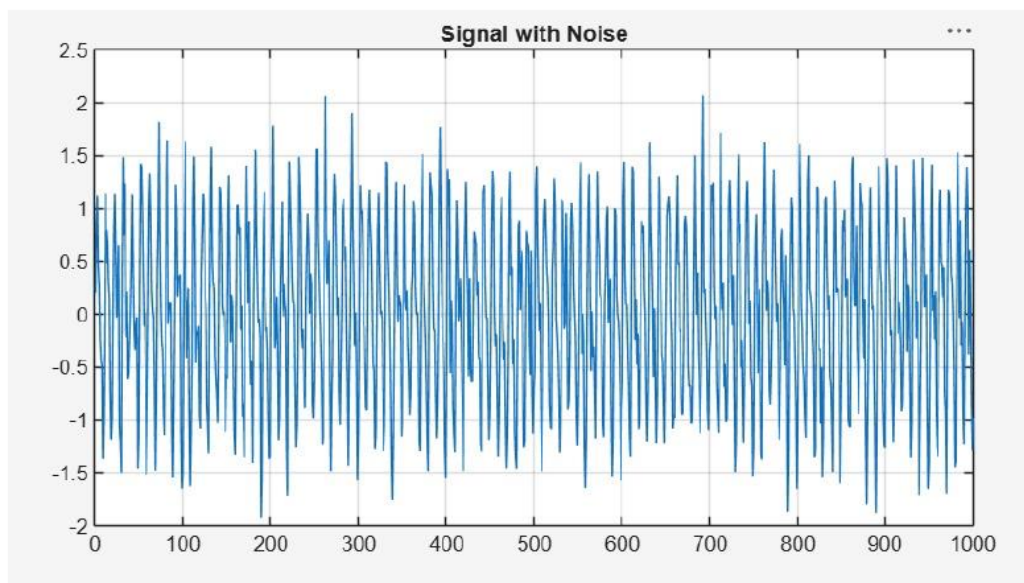
```
title('Signal with Noise')
```

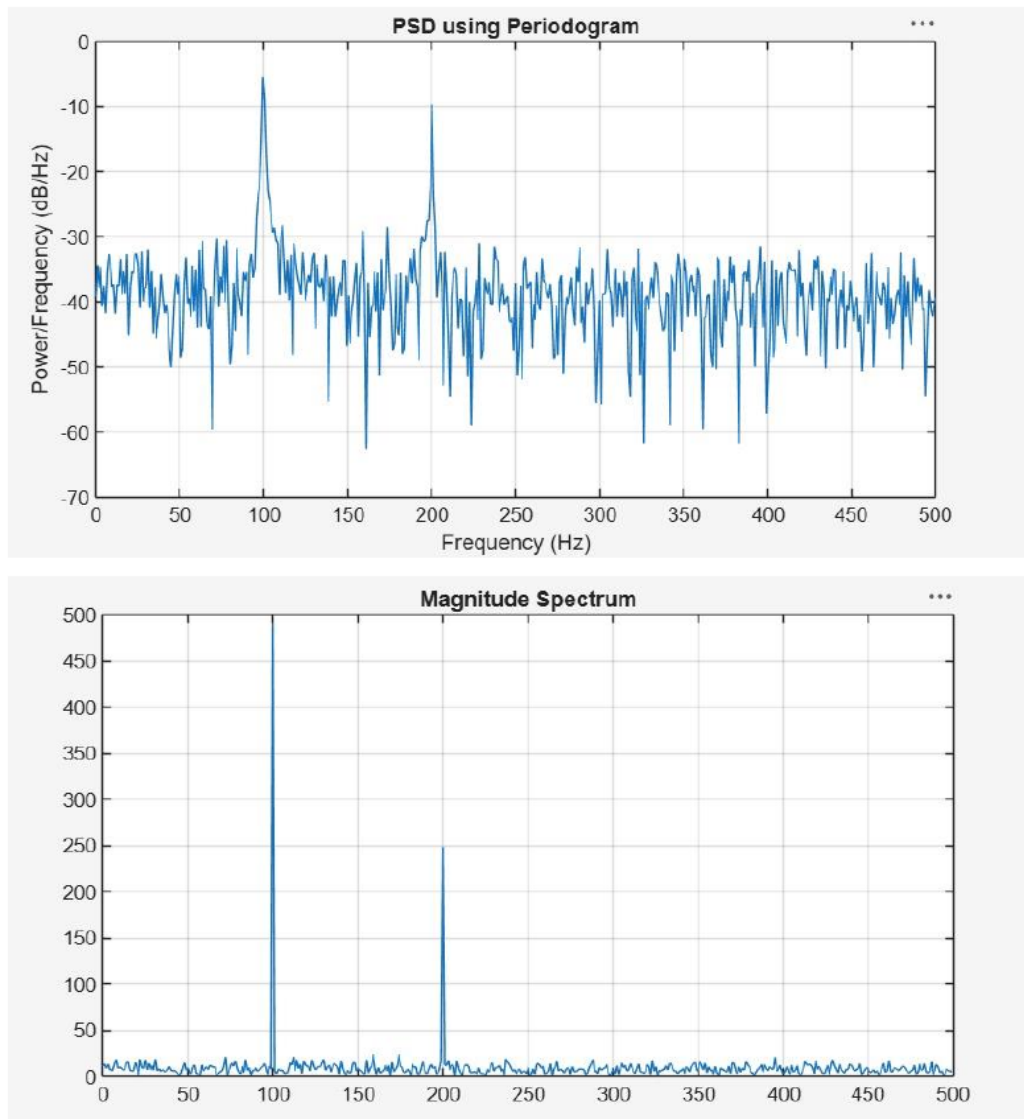
```
figure, periodogram(x,[],[],fs)  
title('PSD using Periodogram')
```

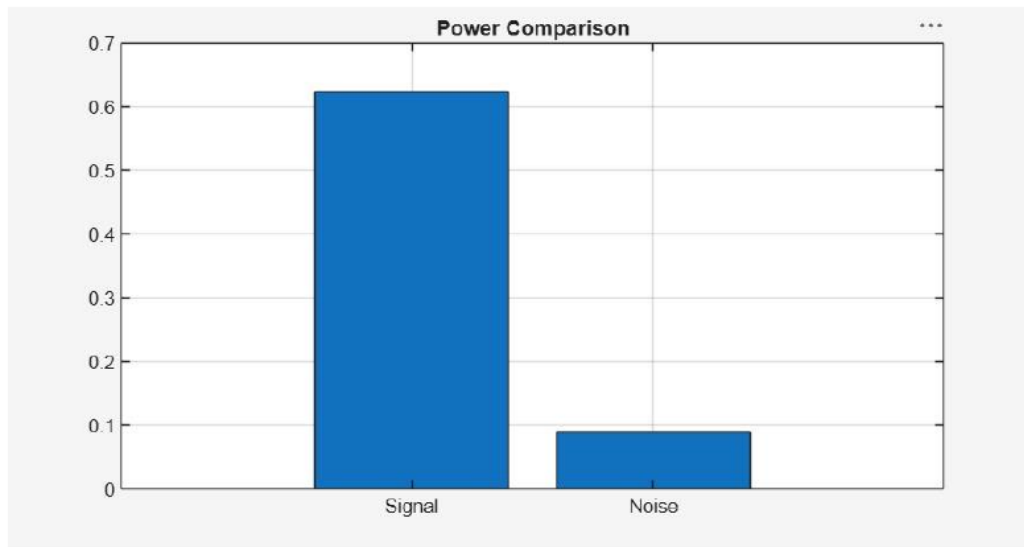
```
figure  
X=abs(fft(x));  
f=(0:length(X)-1)*fs/length(X);  
plot(f(1:500),X(1:500)), grid on  
title('Magnitude Spectrum')
```

```
figure  
bar([mean(s.^2) var(n)])  
set(gca,'XTickLabel',{'Signal','Noise'})  
title('Power Comparison')  
grid on
```

Output:







Result:

1. A discrete-time signal consisting of sinusoidal components with added noise was successfully generated in MATLAB.
2. The Power Spectral Density (PSD) of the signal was estimated using the Periodogram method.
3. The dominant frequency components present in the signal were accurately identified from the PSD plot.
4. The spectral characteristics of the signal were analyzed, showing the distribution of signal power across frequencies and the effect of noise on the spectrum.

Practical Question 1:

Generate a discrete-time signal consisting of two sinusoidal components with frequencies of 100 Hz and 250 Hz, sampled at 1000 Hz. Add white Gaussian noise to the signal and estimate its Power Spectral Density (PSD) using the Periodogram method in MATLAB.

Task:

1. Generate a discrete-time signal containing two sinusoidal components of 100 Hz and 250 Hz sampled at 1000 Hz.
2. Add white Gaussian noise to the generated signal in MATLAB.
3. Estimate the Power Spectral Density (PSD) of the noisy signal using the Periodogram method and plot it.
4. Analyze the PSD plot to identify dominant frequency components and study the effect of noise on the frequency spectrum.

Aim (10)	Program (20)	Procedure/Calculations (20)	Results (20)	Record (10)	Viva (20)	Total (100)

Practical Question 2:

Generate a noisy sinusoidal signal sampled at 2000 Hz and estimate its Power Spectral Density (PSD) using the Periodogram method in MATLAB.

Task:

1. Generate a noisy sinusoidal signal sampled at 2000 Hz in MATLAB.
2. Estimate the Power Spectral Density (PSD) of the signal using the Periodogram method.
3. Plot and analyze the PSD to identify dominant spectral peaks.
4. Determine the frequencies corresponding to the main peaks and evaluate the signal's power distribution across the frequency spectrum.

Aim (10)	Program (20)	Procedure/Calculations (20)	Results (20)	Record (10)	Viva (20)	Total (100)

Expt 7	Spectral Estimation
Date:	7.2 Analysis of Windowing Effects on Spectral Estimation Using Rectangular, Hamming, and Hanning Windows in MATLAB

Objectives:

1. To estimate the spectrum of a signal using different window functions such as Rectangular, Hamming, and Hanning windows.
2. To analyze the effect of window selection on spectral leakage.
3. To compare the main-lobe width and side-lobe levels of different windows.
4. To evaluate the accuracy of frequency component detection.

Procedure:

Result

To estimate PSD of a noisy sinusoidal signal using the periodogram and identify dominant frequency components.

Student Details

Name: Dhinakaran. S

Reg. No.: 192412353

Course: ECA09 - Digital Signal Processing